Consent Gate, Telephony Intake & Audio Normalisation

Case Ace v2.0 Architectural Specification & Compliance Guide
Citizens Advice Wandsworth (CAW) — Advice Quality Standard Level 3 AI Pipeline

1. Executive Summary & Regulatory Foundations

Case Ace v2.0 accepts advice consultations via three distinct routes:

Live In-Person Interview (workstation microphone capture)

Telephone Call via Cisco Webex Calling (WebRTC dual-channel telephony capture)

Historical Recording File Import (sandboxed browser decoding)

Under UK GDPR Articles 6(1)(a), 9(2)(a), and the Citizens Advice Data Protection Policy, recording any client advice consultation requires affirmative, explicit, informed consent. The Consent Gate is the common, non-bypassable architectural entry point across all three routes.

```
flowchart TD
    subgraph Intake Routes
        R1[Route 1: Live In-Person] --> CG[Consent Gate]
        R2[Route 2: Cisco Webex Telephony] --> CG
        R3[Route 3: File Import] --> CG
    end

    subgraph Consent Gate Validation
        CG --> V1{Affirmative Action?}
        V1 -- No --> Block[Intake Blocked / Controls Locked]
        V1 -- Yes --> V2{Zero Client PII?}
        V2 -- No --> Err[Privacy Invariant Thrown]
        V2 -- Yes --> Gen[Generate ConsentRecord]
    end

    subgraph Audio Capture & Ingest
        Gen --> C1[Live Audio Capture]
        Gen --> C2[Webex Stream Capture]
        Gen --> C3[Media Streaming Decoder]
    end

    subgraph Unified Normalisation
        C1 --> Norm[AudioNormalizer]
        C2 --> Norm
        C3 --> Norm
        Norm --> Rep[NormalizedAudioSession<br/>16kHz Mono Float32 PCM<br/>Speaker Channel Map<br/>ConsentRecord]
    end

    Rep --> VolatileStore[(VolatileSessionStore<br/>RAM Only)]
```

2. Phase 6.1: The Universal Consent Gate

2.1 Non-Skip Invariant

Recording cannot start, and file imports cannot be decoded, until the Consent Gate is affirmatively satisfied. There is no bypass, demo, or skip path in the codebase.

2.2 Route-Specific Consent Requirements

2.3 Zero Client PII Invariant

The consent record must never record the client's name, telephone number, National Insurance number, address, or any client identifier:

```
export interface ConsentRecord {
  consentId: string;           // Random UUID (e.g. cst_8f3a...)
  consentedAt: string;         // ISO 8601 timestamp
  route: IntakeRoute;          // 'live_in_person' | 'webex_telephony' | 'file_import'
  adviserId: string;           // Adviser user ID (accountability)
  confirmedByAdviser: true;    // Explicit boolean affirmation
  originalAppointmentDate?: string; // YYYY-MM-DD (Imports only)
  importConsentMeans?: string;      // Method description (Imports only)
}
```

[!IMPORTANT]
Audit Trail Invariant: The link between the consent record and the client belongs strictly in Casebook, in the adviser's own case record notes, not in Case Ace.

2.4 Immediate One-Action Consent Withdrawal

Under UK GDPR Article 7(3), a client may withdraw consent at any time. When a client withdraws consent:

Activating the visible, always-accessible "Withdraw Consent (Instant Destroy)" button destroys the session immediately.

Zero Confirmation Dialogues: There is no "Are you sure?" modal delay.

All volatile session memory is wiped, recovery workers are terminated, and ArrayBuffer instances are zero-filled.

Webex Telephony Call Continuity: On a Webex call, consent withdrawal immediately ceases recording and destroys session data, but leaves the telephone call active so the adviser and client may continue their consultation unrecorded.

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3. Phase 6.2: Route 1 Live Capture & Quality Controls

3.1 Web Audio Capture Architecture

Sample Rate: 16,000 Hz (16 kHz).

Channels: 1 (Mono Float32 PCM).

Storage: Streamed in 4096-sample buffer chunks directly into volatile heap memory.

3.2 Persistent Recording Indicator

Sticky Top Bar: Positioned at top: 0 with z-index: 1000, visible across all scroll positions.

Flashing Badge: Pulsing red indicator %Ï REC with active elapsed timer (MM:SS).

Browser Tab Notification: Synchronizes document title with [%Ï REC] Case Ace - Live Consultation.

3.3 Volatile Memory Pressure Monitoring

To prevent out-of-memory crashes on CAW's 8GB RAM laptops, memory consumption is monitored in real-time (Float32 = 4 bytes/sample = ~3.84 MB/min):

Normal (< 45 min / < 172.8 MB): Green indicator.

Moderate (45 – 60 min / 172.8 – 230.4 MB): Amber advisory.

High Memory Pressure (60 – 90 min / 230.4 – 345.6 MB): Red alert recommending interview conclusion.

Quota Limit (>= 90 min): Hard stop preventing uncontrolled allocation.

3.4 Single Dominant Speaker Detection

Clinical Rationale: If only the adviser's workstation microphone is active and the client is seated across the room or on speakerphone, the microphone may capture only the adviser's voice. A transcript missing client instructions will cause the drafting model to misstate what the client instructed.

Acoustic Algorithm:

Computes RMS energy across 100ms analysis frames.

Dynamically estimates noise floor and partitions voiced frames into high-energy (close-mic) vs secondary-energy tiers.

Evaluates turn-taking frequency.

If one acoustic tier accounts for $\ge 88%$ of voiced speech or if fewer than 2 conversational turns occur across $>20\text{s}$ of speech, surfaces an on-screen warning:

& þ Single Dominant Speaker Detected: Single dominant speaker detected (94% speech dominance). Ensure the client's voice is clearly audible near the microphone so their instructions are accurately recorded.

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4. Phase 6.2: Route 2 Cisco Webex Telephony Stream Capture

Gated Recording Control: The "Start Recording Call" button is strictly locked until the adviser verifies client consent on the call.

Speaker Channel Splitting:

Channel 0: Local adviser microphone.

Channel 1: Incoming remote client WebRTC telephone audio.

Decoupled Telephony Lifecycle: Allows independent control of audio recording vs telephony call stream.

5. Phase 6.3: Audio Normalisation

5.1 Universal Normalized Representation

All three intake routes converge on NormalizedAudioSession:

Format: FLOAT32_PCM_16KHZ_MONO

Sample Rate: 16,000 Hz

Channels: 1

Speaker Map: { isDualChannel: boolean, adviserChannel?: number, clientChannel?: number, sourceType: ... }

Consent Record: Immutable ConsentRecord

5.2 Downstream Intake Agnosticism

Phases 7 through 14 (Pass 1 acoustic redaction, Speech-to-Text v2, tokenisation, LLM drafting, and Casebook clipboard handoff) consume only NormalizedAudioSession and have zero route-specific logic.

5.3 Non-Sensitive Intake Telemetry

To allow CAW quality assessors to measure whether note quality differs by intake route (e.g. comparing in-person vs imported audio), the frontend sends non-sensitive telemetry:

```json
{
  "stage": "intake_completed",
  "intakeRoute": "webex_telephony",
  "durationMs": 184500,
  "success": true
}
```

Telemetries are scrubbed of all PII, client names, phone numbers, and session text.